

MARKUS KUNEJ

Toronto, ON

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EXPERIENCE

Instacart

Apr 2024 – Jun 2026

Software Engineer — ML Platform

Toronto, ON

Core contributor to Griffin, Instacart's ML platform for training, batch inference, and online serving supporting TensorFlow, PyTorch, LightGBM, and ONNX models; used by Ads, Economics, Fraud Detection, and Batching/Logistics teams.

- Led the **TensorFlow 2.8 to 2.15 migration** across training, batch inference, serving, and the Griffin SDK; coordinated rollout with approximately 8 downstream teams and reduced average serving prediction-step latency by **5-10%**.
- Owned and shipped a production **Arize integration** using Temporal workflows to orchestrate training-data uploads and per-model dashboard creation, enabling live drift and accuracy monitoring alongside production inference results.
- Owned and shipped **ML scorecards** across a Django backend and React UI, using Temporal workflows to schedule and execute evaluations; assigned production models bronze, silver, or gold status based on dataset size, feature count, monitoring coverage, and model age.
- Expanded **distributed LightGBM training and batch inference** with coordinated early stopping, multiclass predictions, and labeled Snowflake outputs; added preprocessing artifact support and non-model column pass-through.
- Maintained and extended the **Griffin Python SDK** for ML engineers using code-based workflows, ensuring feature parity with the platform UI and managing optional ML framework dependencies and version constraints to minimize conflicts with teams' existing codebases.
- Participated in the **ML Platform on-call rotation**, investigating and mitigating production issues spanning online model-serving latency, feature-store data retrieval, and model training failures.
- Created project-local **Claude Code skills and plugins** for Griffin training-job creation and workflow debugging; added repository guidance to support agent-assisted development.

Huawei Technologies Canada

May 2021 – Jun 2022

Software Engineer Intern, Distributed Systems and AI

Markham, ON (Remote)

- Developed CPU-to-GPU and AI accelerator offloading for dataset operations in MindSpore, an open-source ML framework, reducing training time by **20%** on networks including ResNet and AlexNet.
- Implemented CPU, GPU, and AI accelerator kernels using C++, CUDA, and Python; profiled training loss, accuracy, and memory use to identify performance bottlenecks.

Untether AI

May 2020 – Sep 2020

Hardware Engineer Intern

Toronto, ON

- Built Python test automation for JTAG and RTL verification of an AI accelerator using FPGA hardware, cocotb, and Synopsys VCS; wrote C firmware to control accelerator-card GPIO and voltages.

TECHNICAL SKILLS

Languages: Python, SQL, Go

ML & Data: TensorFlow, PyTorch, LightGBM, Ray, Arize, Snowflake

Backend & Infrastructure: Django, AWS (S3, IAM), Kubernetes, Docker, Terraform, Temporal, Protobuf

Frontend: React

Developer Tools: Claude Code (custom skills, plugins, repository guidance)

EDUCATION

University of Toronto

Sep 2018 – May 2023

B.A.Sc. in Engineering Science, Machine Intelligence Major

Toronto, ON

Business Certificate | Schulich Leader Scholarship (\$100,000), 2018